Paper Review • arXiv 2611.05377 • 2026-11-03

DreamX

Video Diffusion: A Universal LAM

Video Diffusion Model is a Universal Latent Action Learner

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Source: paper, Figure 1

TL;DR: Video diffusion reads out actions, then acts out the future

DreamX uses pretrained video diffusion (Cosmos-Predict2.5 2B) **as both encoder and decoder**: queries read compact latents from spatiotemporal features; future video is decoded from only **“first frame + latents”** . The prior fills in appearance; latents keep motion—a shared human–robot action interface.

300×

Latent action token compression

1,200 video tokens per latent frame → 4 latent action tokens

0.17 | 69.7%

LARYBench Regr. MSE | Cls. Acc.

Prev. best DiLA: 0.38 | 62.25%

96.7%

Real-robot drawer-close subgoal SR

Mimic-Video 50%; 1.47 s per inference, 19.1× faster

Embodiment-bound labels; appearance-entangled latents

① Labels not shareable

Human videos lack robot action labels

Robots differ in control spaces

Same behavior ≠ same low-level action

② Appearance entangled

LAMs train the encoder by "reconstructing future frames"

Viewpoint and scene appearance leak into latent actions whenever they aid reconstruction

Limits cross-embodiment transfer

③ Too narrow a view

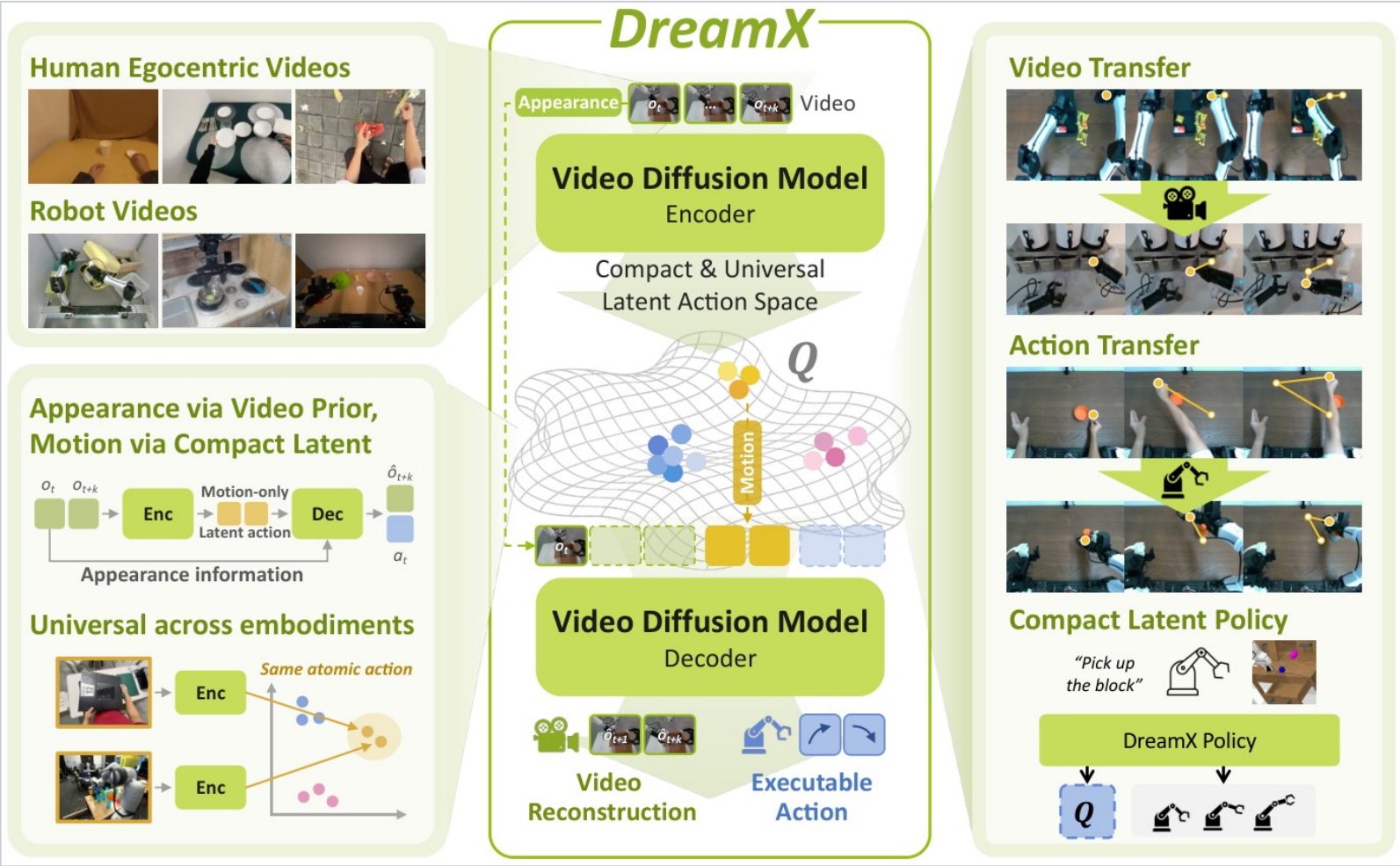
LAPA / Genie / UniVLA: learned from scratch on frame pairs only

AdaWorld / VideoWorld2: video model is only a generator; action encoder still external

Key Insight

Pretrained video diffusion models carry two priors: **spatiotemporal representations** (motion / correspondence / geometry / semantics) to "read" actions; **generative priors** (synthesizing plausible futures from a first frame) to "supervise" them.

Core idea: video prior → appearance; compact latent actions → motion



Training

- Egocentric human + robot videos, no pairing needed

Encoding

- Video diffusion model → compact cross-embodiment latent action space Q

Decoding

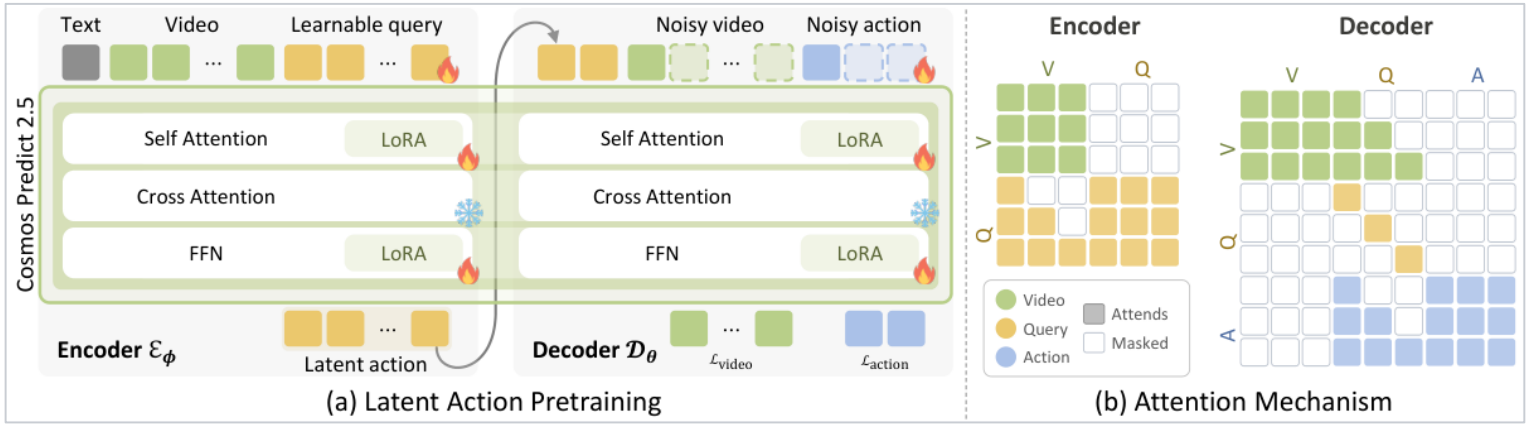
- First frame gives appearance, Q gives motion → reconstructed video / executable actions

Applications

- Video transfer • action transfer
- compact latent policy

Source: Figure 1

One backbone encodes and decodes; attention masks route information



Encoder E_ϕ

- 4 learnable queries per latent frame
- Share spatiotemporal attention with video tokens
- Frame-causal query→video: Q_g sees only V_0:g

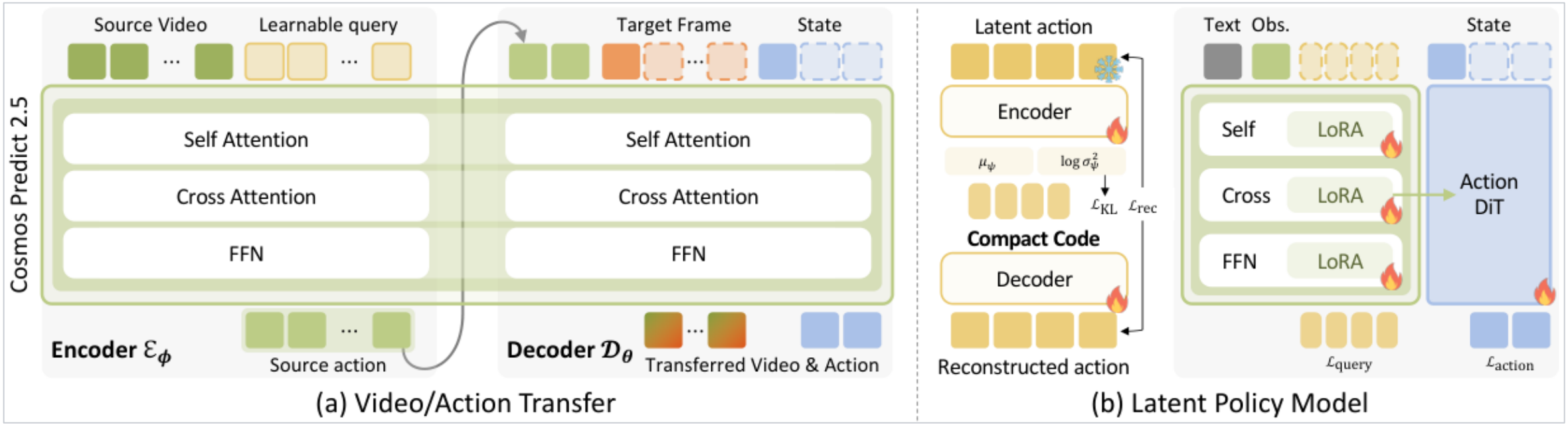
Decoder D_θ

- Conditioned only on first frame V_0 + latent actions Q (no text)
- With labels, adds state p_0 to aid action decoding
- Video and action tokens mutually masked; read only Q

L = L_video + λ • L_action flow matching; λ=0 without action labels; backbone frozen

Source: Figure 2, § 3.2, Appendix A.2

Three uses of one latent space: video/action transfer, latent policy



(a) Motion transfer: no extra training

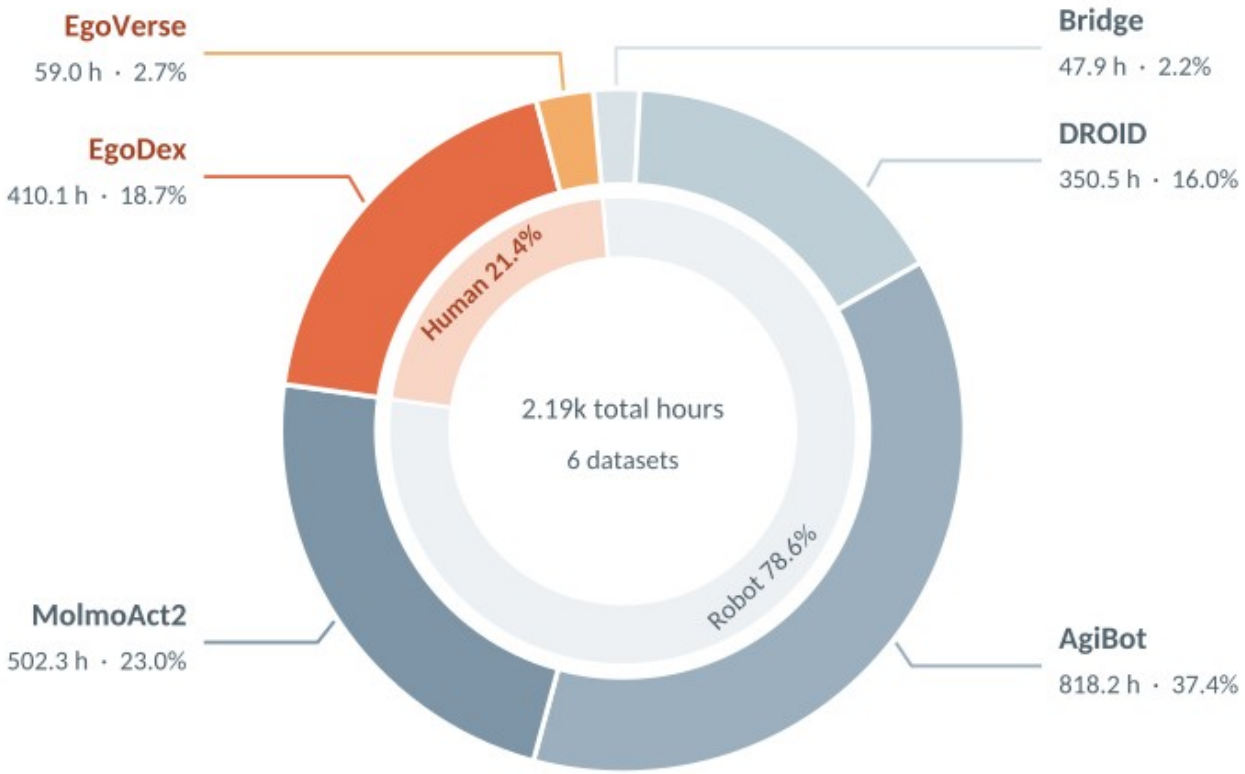
$\hat{V}^{tgt} = D(Q^{src}, V_0^{tgt})$ $\hat{A}^{tgt} = D(Q^{src}, p_0^{tgt})$

(b) Latent policy: $\pi(Z, A \mid V_0, \ell, p_0)$

FAE compresses each token 2048 → 32 dims; human videos supervise only Z, robot data also A

Source: Figure 3, § 3.3–3.4, Appendix A.3

2.19k hours of heterogeneous data, ~1/5 from egocentric human video



2.19k h

Total hours

21.4%

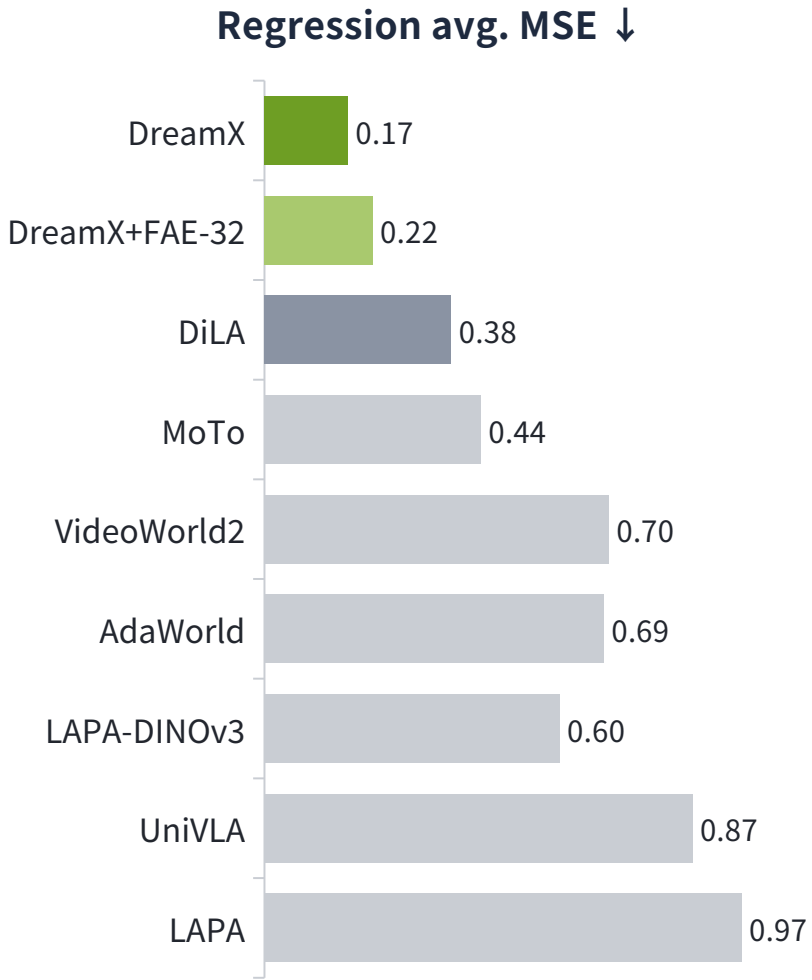
Human video

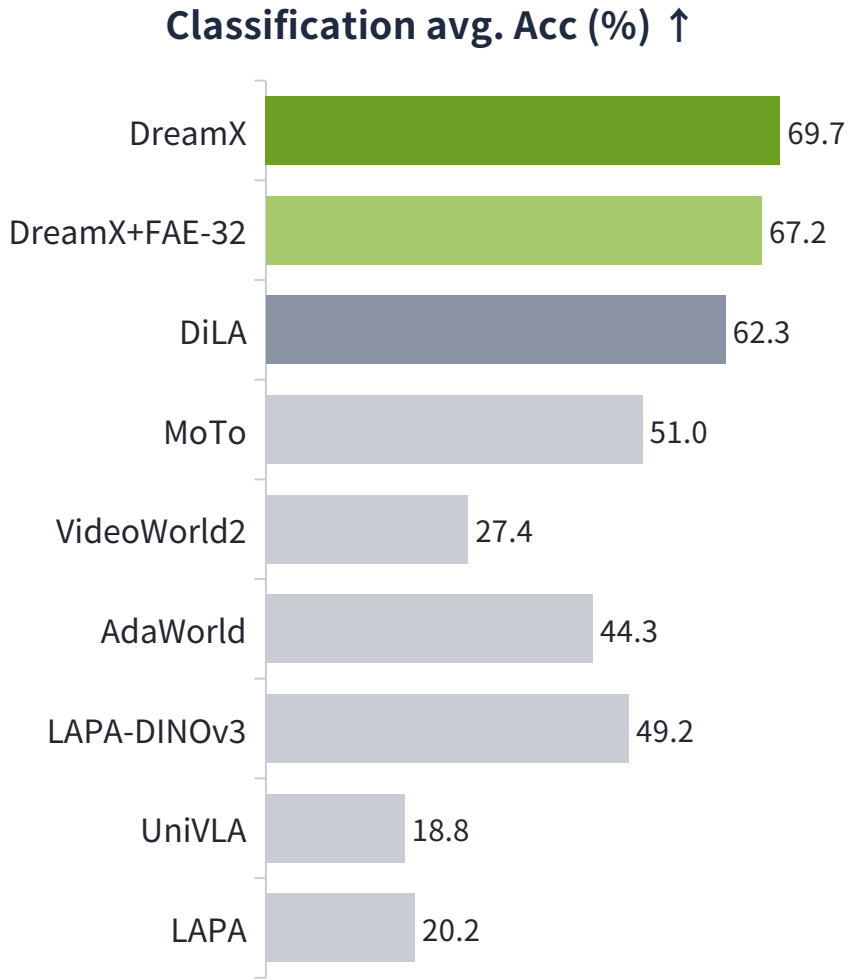
Key Settings

- Backbone: Cosmos-Predict2.5 2B, frozen weights
- Clips: 5–29 frames → 2–8 latent frames, ⩽32 queries
- Separate rank-16 LoRA for encoder and decoder
- LR 1e-4 (LoRA, queries), 78,000 steps
- Equal video and action loss weights when labels exist

Source: Figure 9, Appendix A.1–A.2

LARYBench: regression error more than halved, accuracy up 7.4 points





0.38 → 0.17

Regression MSE (vs. DiLA)

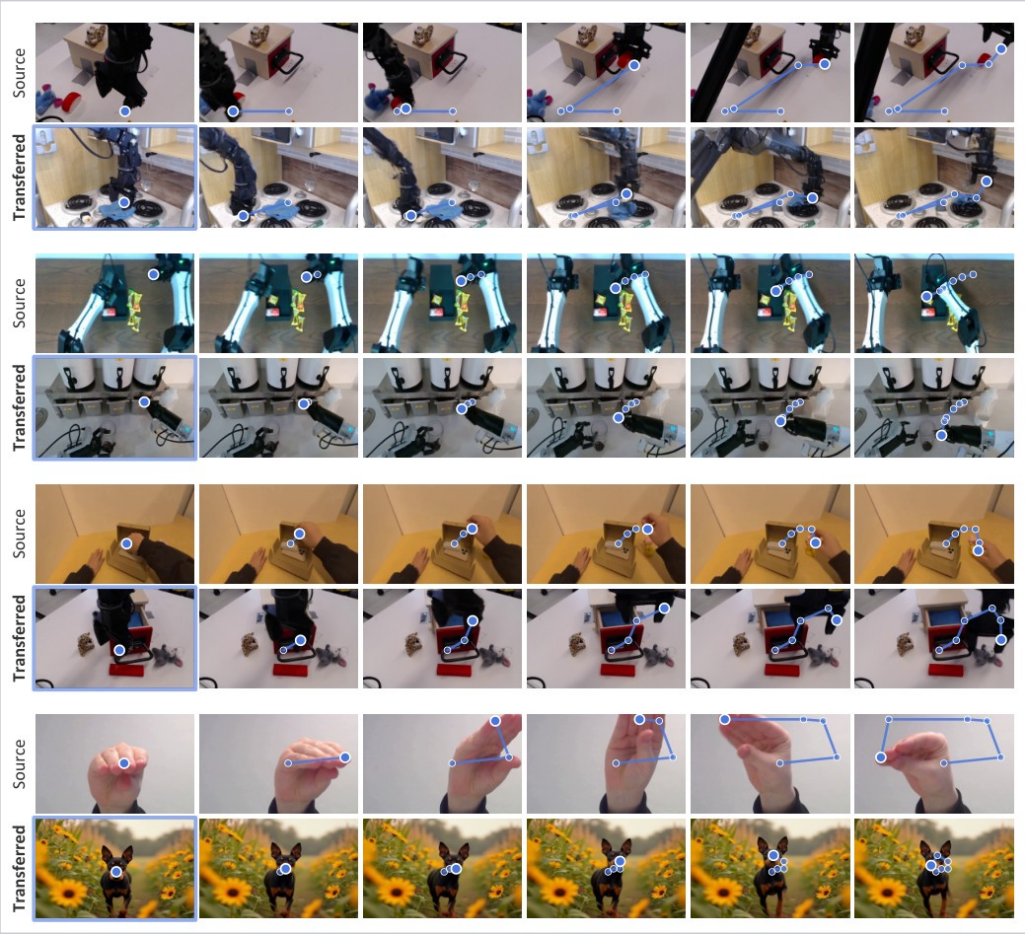
62.3 → 69.7

Classification acc. % (vs. DiLA)

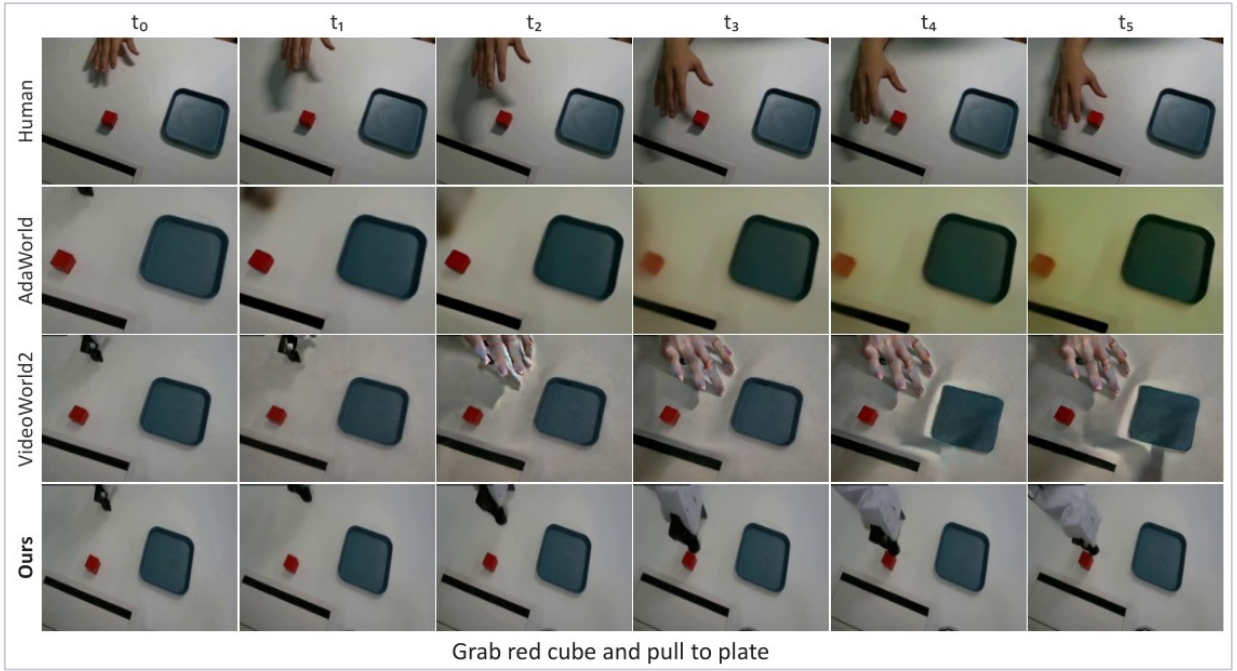
- FAE-32, compressed to 32 dims, still beats all baselines
- AdaWorld / VideoWorld2, also using video priors, lag behind: the key is “reading the backbone’s internal features directly”

Source: Table 1 (frozen features + attentive probe)

Video transfer across scenes/embodiments, no hand “leak”



Any-to-any transfer: robot↔robot, human→robot, human→DAVIS
(blue dots: end-effector trajectory)



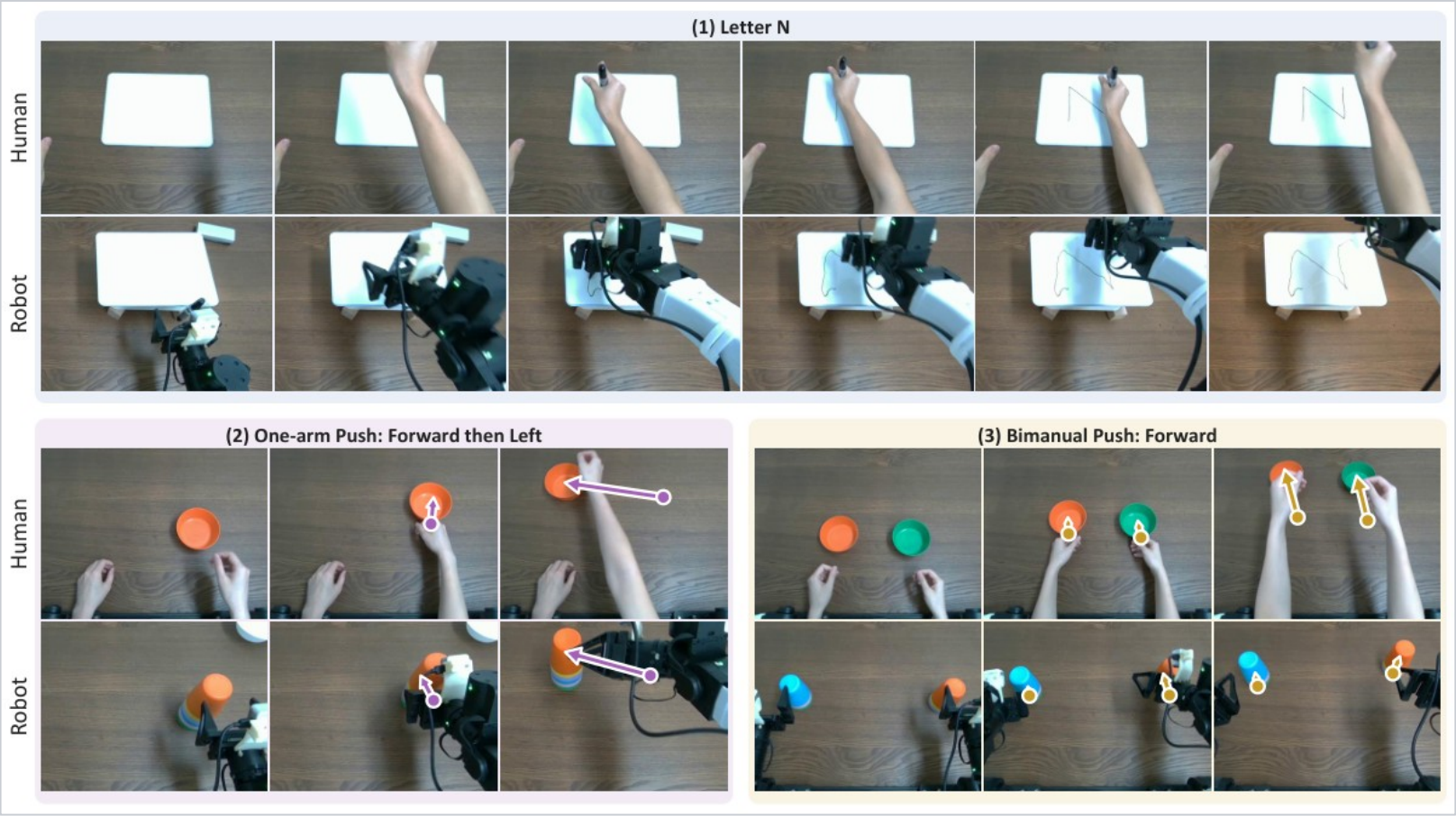
HumanAndRobot zero-shot human→robot (Grab red cube and pull to plate)

No paired human–robot data used in training; in Table 2, FVD / LPIPS / PSNR / SSIM all beat AdaWorld and VideoWorld 2

AdaWorld: degraded frames VideoWorld 2: occasional human hands
DreamX: robot appearance preserved

Source: Figure 4, Figure 5, Table 2, Appendix B.5

Action transfer: human video → robot joints, trajectories kept open-loop



Writing letter “N”

Fine spatial trajectory + stroke order

One-arm push: fwd→back→left

Direction changes + temporal order

Two-arm push: forward

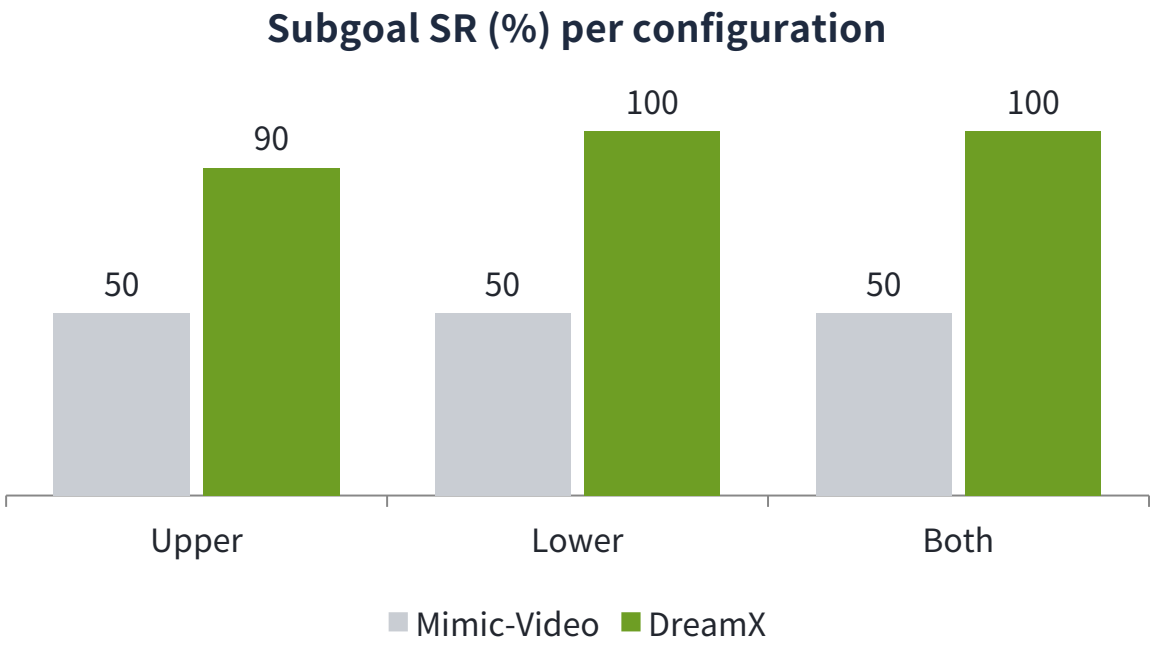
Each arm’s motion mapped separately

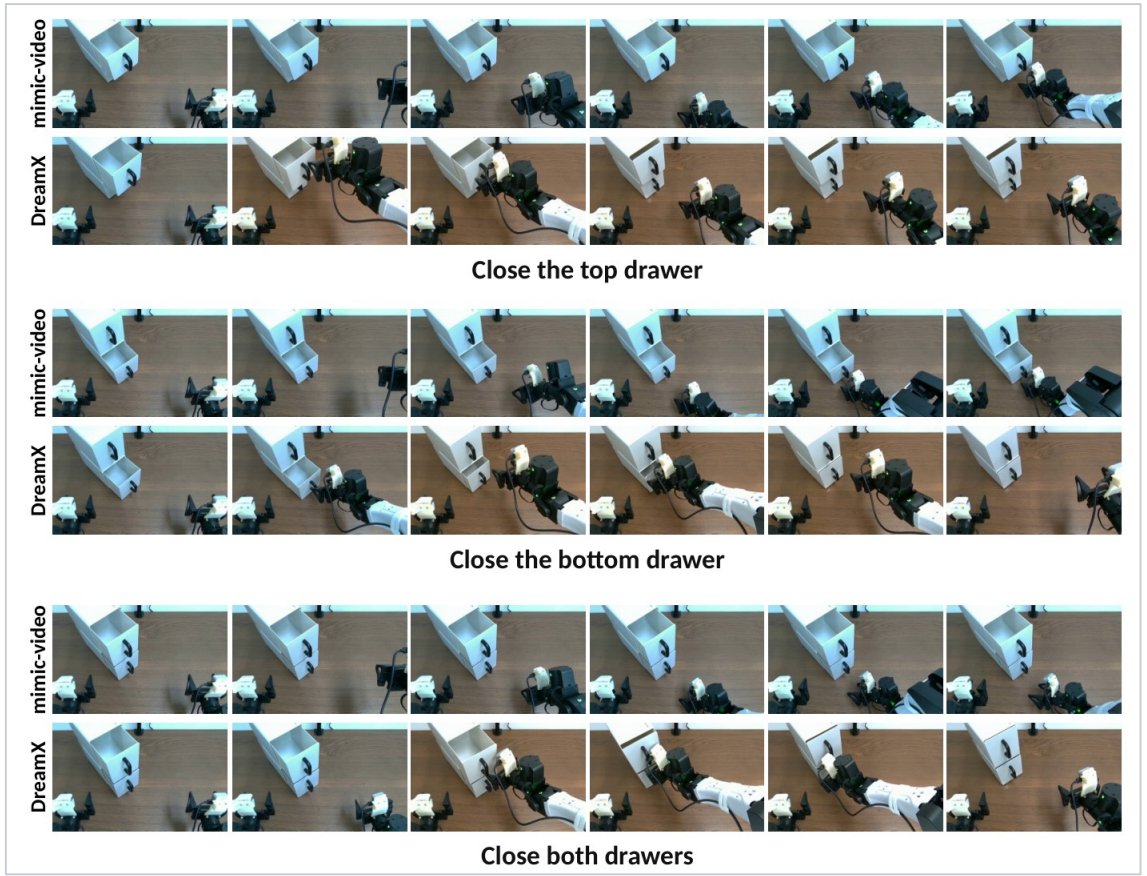
Setup

Bimanual YAM, 14-D absolute commands; no visual feedback or paired data. Qualitative only; no quantitative metrics reported.

Source: Figure 6, § 4.3, Appendix B.1–B.2

Latent policy: 96.7% SR closing drawers, 19.1× faster than video policy





96.7%
DreamX avg. SR

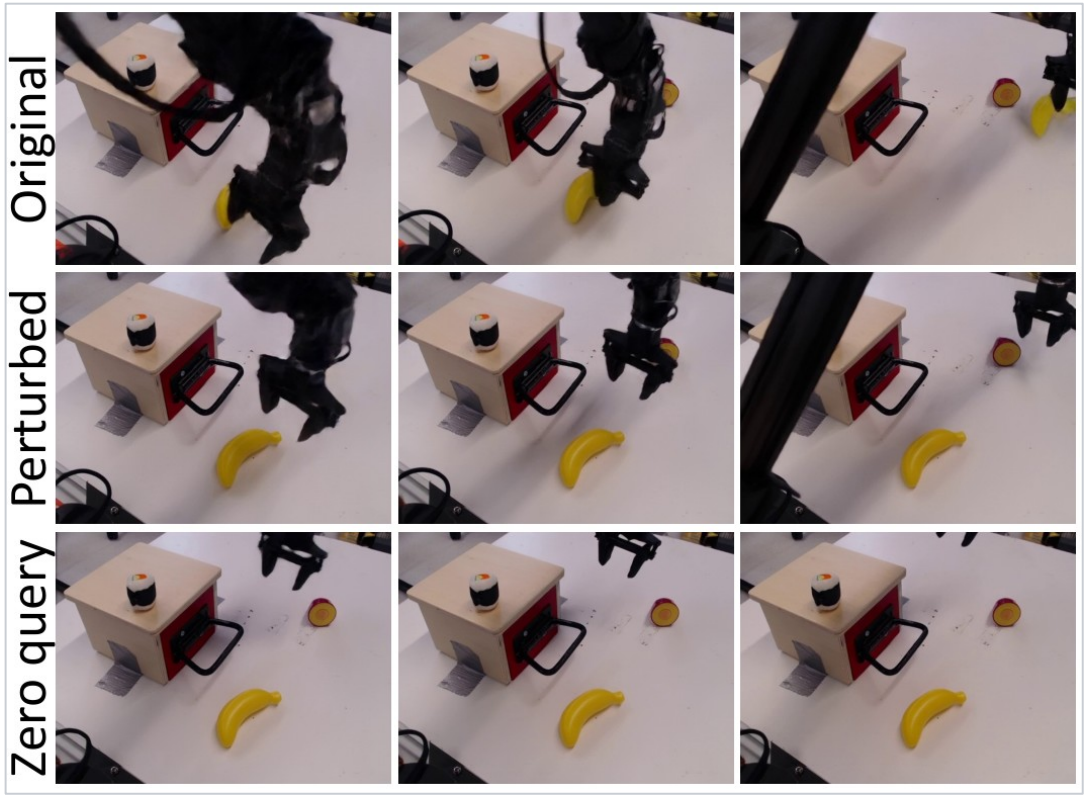
50%
Mimic-Video

1.47 s
19.1× faster/call

Human video only: upper-drawer setup (no robot action labels) still reaches 100% (App. Table 6)

Source: Tables 3, 5, 6, Figure 19, Appendix B.3

Latents drive motion, not appearance: zeroed ≈ motionless

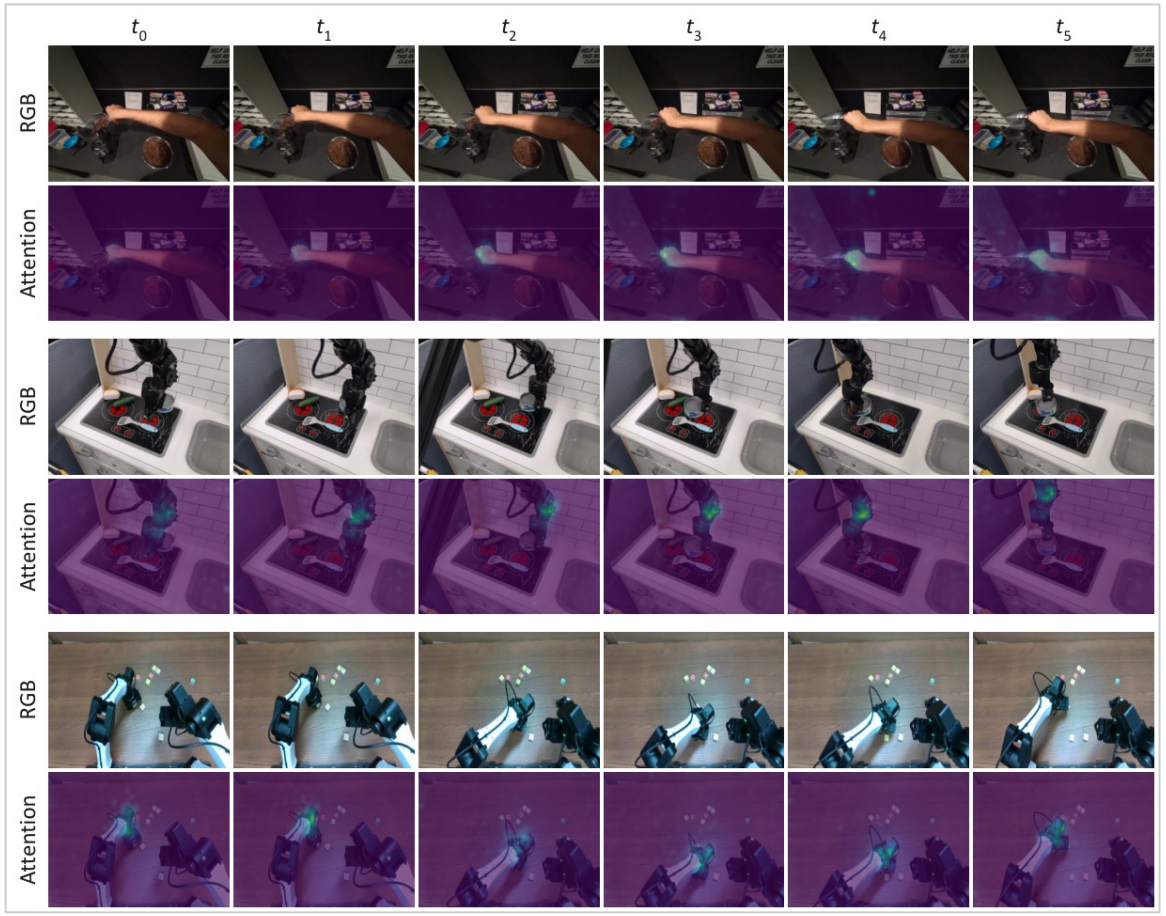


Conditioning	Motion ratio	Flow dev.	RGB L1	Static L1
Original	1.000	0.000	0.0000	0.0000
Noise α=0.1	1.000	0.141	0.0054	0.0030
Noise α=0.3	0.995	0.211	0.0077	0.0034
Noise α=1.0	0.946	0.481	0.0215	0.0077
Zero queries	**0.456**	**0.866**	**0.0758**	**0.0227**

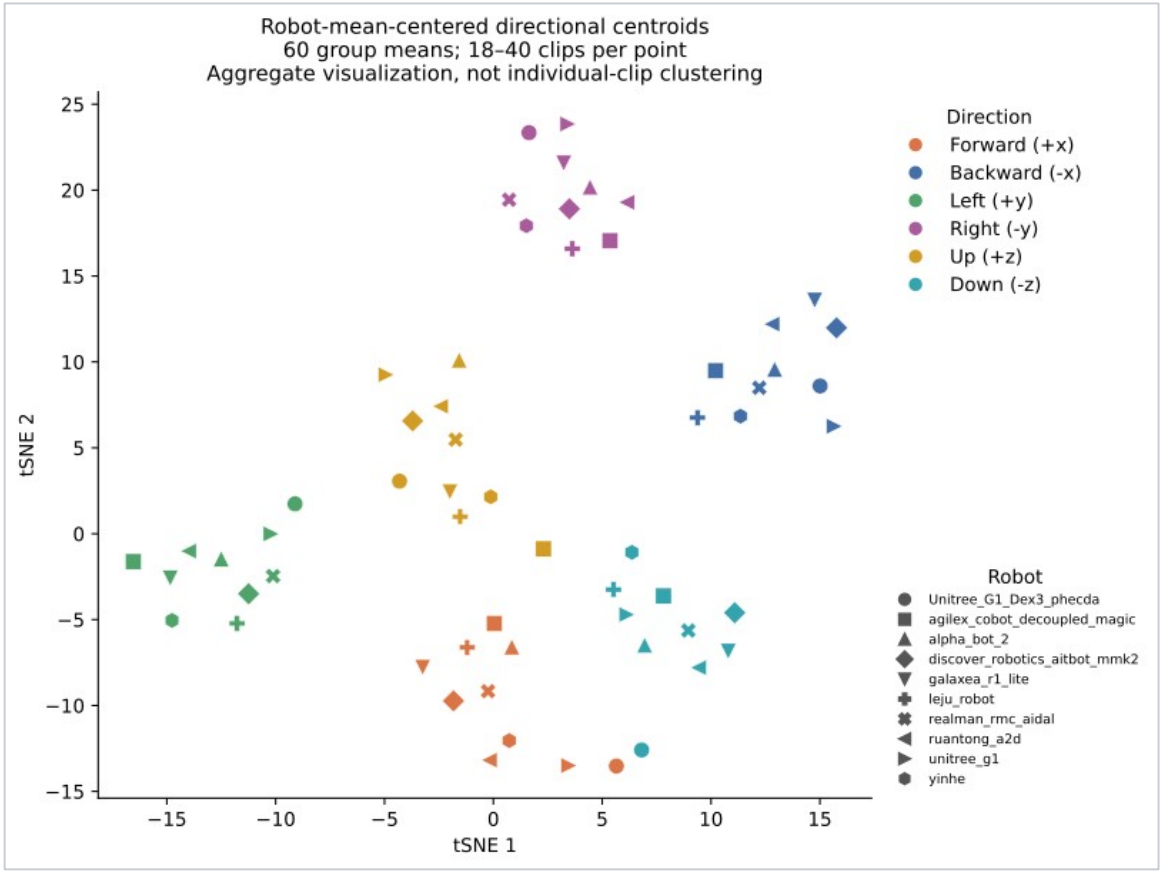
- Stronger perturbation → larger deviation in decoded motion
- Zeroing leaves a motion ratio of only 0.456
- RGB change in static regions (0.0227) ≪ full frame (0.0758) → appearance mostly kept by first frame and generative prior

Source: Figure 7, Table 8, Appendix C.3

Latents track hands/grippers; robot-shared directions



Query→video attention focuses on moving hands / grippers and manipulated objects



RoboCOIN: centroids of 10 robots × 6 directions cluster by direction, not robot

Source: Figure 8, Figure 12, Appendix C.3, D.4

Ablation: labels aren’ t key; latent actions do help predict actions

Aux. action supervision (Table 4, subset)

Supervision	Reg. avg. ↓	Cls. avg. ↑
RGB only	0.24	65.3
RGB + action	0.23	64.8

CALVIN: action head ± latent actions (Table 7)

Policy	⩾1	⩾2	⩾3	⩾4	⩾5	Avg. Len.
w/o latent actions	83.4	61.7	42.5	29.8	20.4	2.378
DreamX policy	**85.7**	**67.0**	**49.5**	**36.4**	**25.7**	**2.643**

≈ On par

Adding action labels: regression −0.01, classification −0.5 pts

→ Motion and action semantics come mainly from video itself

2.378 → 2.643

Avg. completed length; success rates up at all five length thresholds

→ Latent actions effectively guide low-level action prediction

Source: Tables 4 and 7 (Appendix C.2)

Simple and effective; real-robot and fairness evidence still limited

Strengths

- Clean design: one video backbone both "reads features" and "supervises generatively", adding only queries + masks + LoRA
- Interpretable disentanglement: first frame and generative prior handle appearance, backed by perturbation tests and no hand "leakage"
- One representation, three uses: video transfer, open-loop action transfer, latent policy
- Human videos can directly supervise the policy's latent-action branch

Limitations / Concerns

- "300×" counts tokens only; each token is still 2048-D, so policies need an extra FAE
- LARYBench comparison: 2B backbone features + language instructions; pretraining includes AgiBot, possibly overlapping AgiBotWorld-Beta
- Small real-robot study: one drawer-closing task, 10 trials per setup, one baseline; no quantitative action-transfer metrics
- Auxiliary action supervision adds little; 1.47 s per inference is still slow for high-frequency closed loop

Source: reviewer's analysis of the paper and appendix

Takeaways

01 Video foundation models are action representation learners

Reading latent actions inside a pretrained video diffusion backbone far outperforms learning from frame pairs from scratch (LARYBench 0.38 → 0.17 MSE)

02 The generative prior is a natural disentangler

The decoder sees only the first frame + latent actions; the prior fills in appearance, forcing latent actions to encode only motion

03 A shared action interface bridges humans and robots

Without paired data: video transfer, open-loop action transfer, and latent action policies supervised by human video

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